LECTURE 1

# REPRESENTATIONS

## What is a representation?

PRESS → OR SPACE.

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# Learning outcomes

**Discuss** representations of the same object and **choose** a suitable one for a problem.

Not comfortable — Very comfortable

Click to respond

**Discuss** the similarities and differences in visual, word, or geospatial representations.

Not comfortable — Very comfortable

Click to respond

**Understand** how cosine similarity measures how close two embeddings are.

Not comfortable — Very comfortable

Click to respond

**Understand** how PCA and t-SNE visualize high-dimensional vector representations.

Not comfortable — Very comfortable

Click to respond

**Describe** how tokenization, embeddings, and next-token prediction represent language.

Not comfortable — Very comfortable

Click to respond

**Understand** what geospatial information is encoded in geospatial embeddings.

Not comfortable — Very comfortable

Click to respond

# This Lecture: Ideas and Methods

### Philosophy

**What do we mean by a representation?**

- How does our mind represent things and places?
- Encoding vs. meaning
- Re-thinking ideas we take for granted

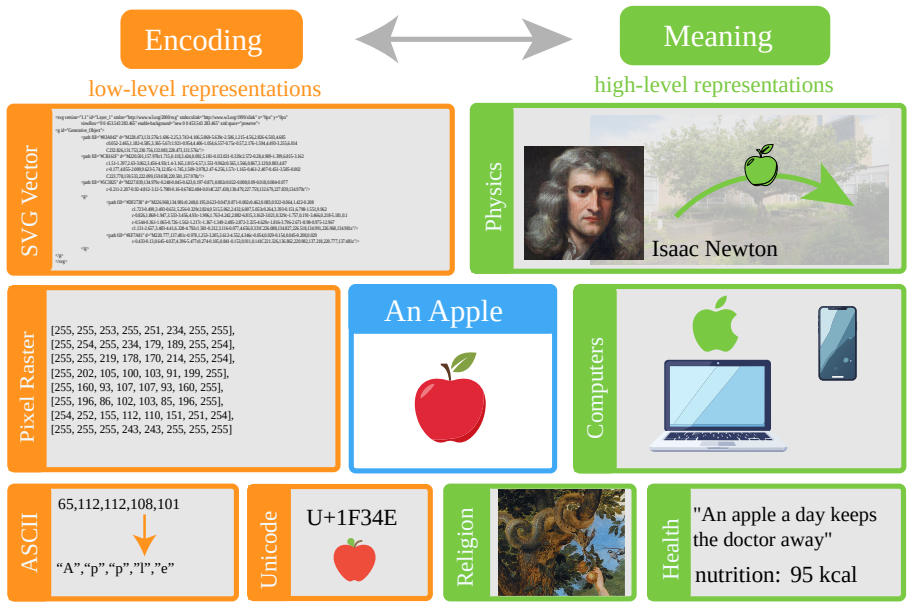
### Technology

**How do computers represent the world?**

- Computer vision
- Natural language processing
- Math: vectors and similarity
- Methods: PCA, t-SNE, embeddings

# Describe an Apple.

## Meaning and Encoding of Representations



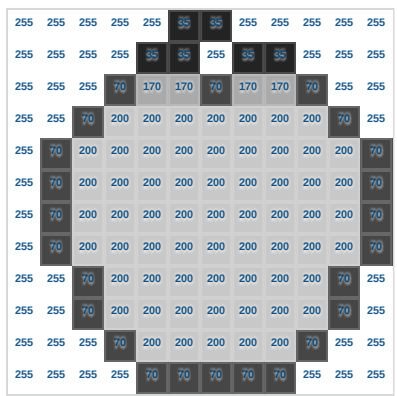

**The same object can be encoded in many ways. A good encoding representation makes useful structure visible and computation easier.**

**But representation is not only encoding.**
**Representation is also meaning.**

# Choosing Right Encoding Representations

## Examples

### Images




### Math

XLVII + LXXVIII = ?

47 + 78 = 125

### Places

COORDINATES

50°43'57.73" N,
7°06'16.63" E

PLACE

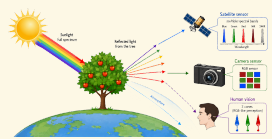
**Hofgarten, Bonn, Germany**

# Lecture Outline

## Representations

What are representations?

## Vision



How vision encodes the world

Die Stadt in meinem Kopf. (2016). *Süddeutsche Zeitung*.

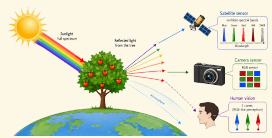
# Lecture Outline

## What is a representation?

### Representations

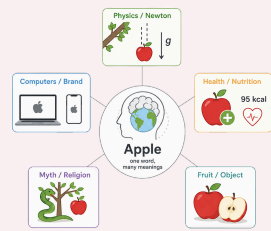
What are representations?

### Vision



How vision encodes the world

### Language




How text becomes meaning

### Spatial

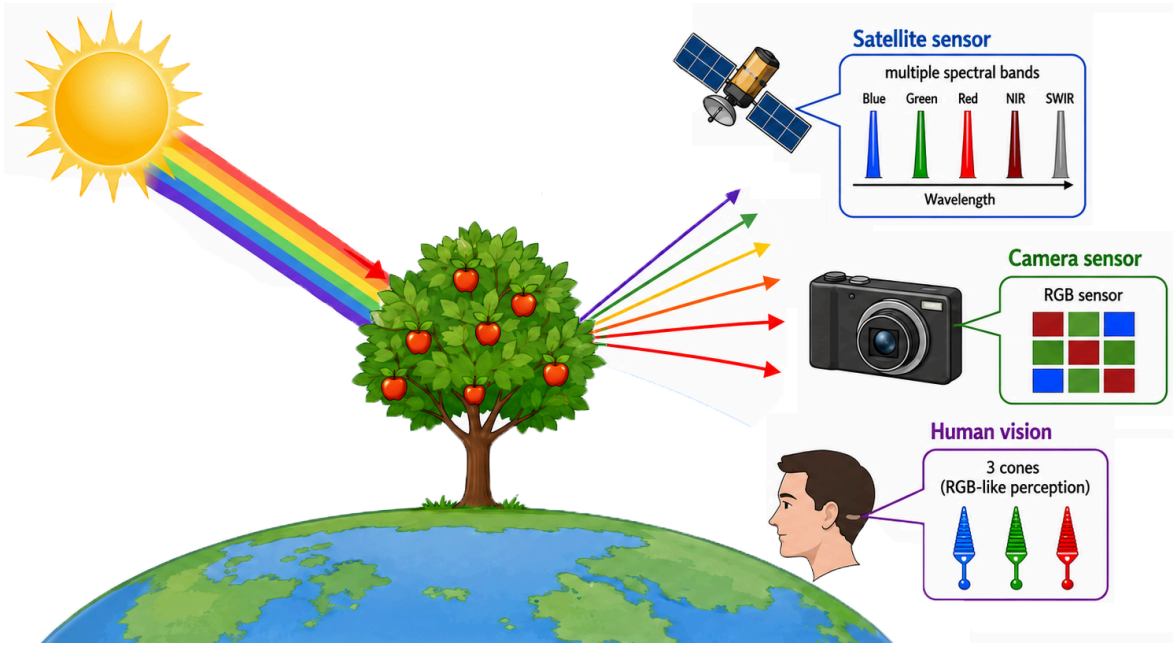
How places carry spatial meaning

Die Stadt in meinem Kopf. (2016). *Süddeutsche Zeitung*.

# Visual Representations

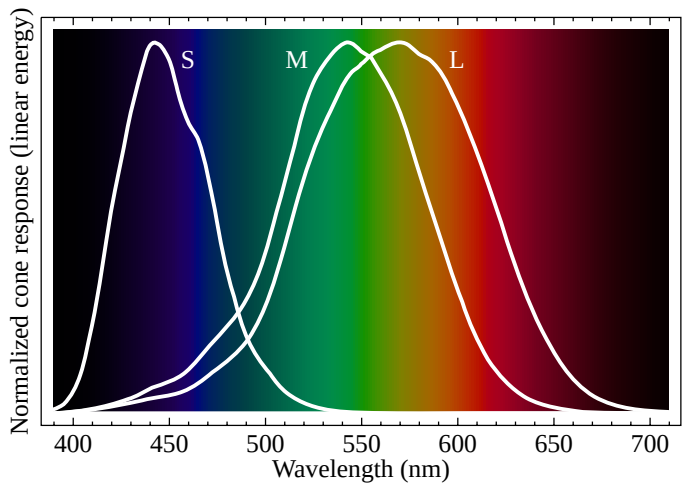
# We perceive our world in representations

**Example: Vision**

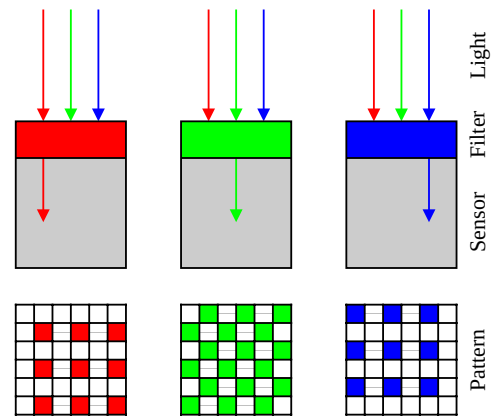


# Engineering suitable Representations.

### Sensitivity of cone cells (S, M, L) in human eyes

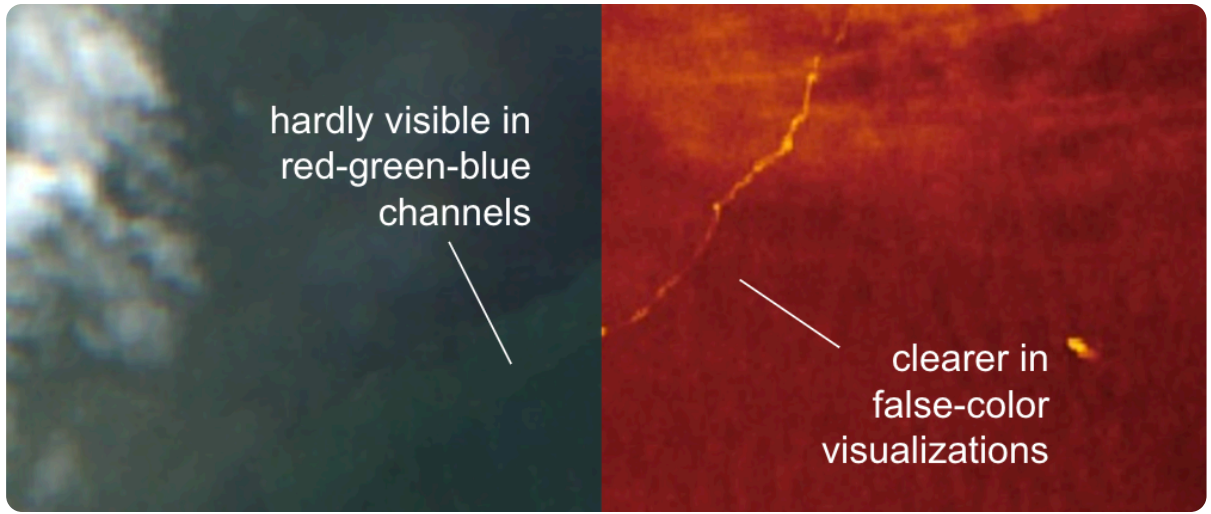



### Bayer Filter on Cameras




**Cameras are built to mimic our eyes: More green pixels in CCD camera mimics the color sensitivity in our eyes.**

# Spectral information can make patterns visible.




Example: plastic litter and marine debris (Sentinel-2)

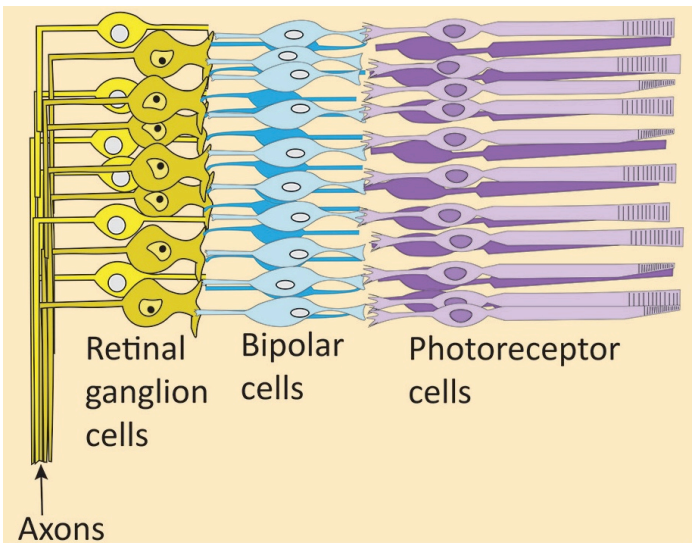
# The representation depends on the Problem

**A useful representation depends on the problem we are trying to solve.**

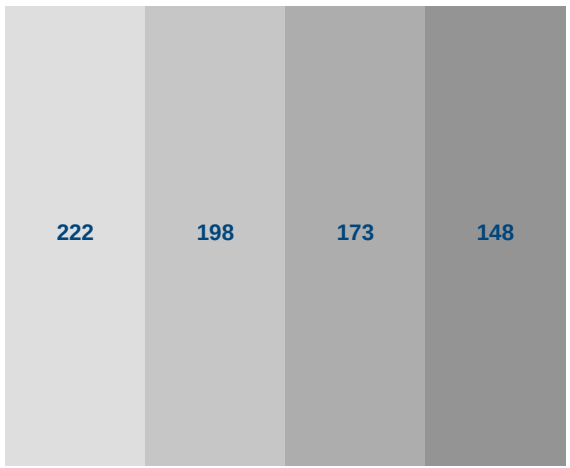
**Bees see ultraviolet patterns that humans miss, because their visual system is tuned to different tasks.**

# Our eyes have edge detectors

**Retinal cells implement an edge filter through lateral inhibition.**




**Mach Bands - Optical Illusion (Mach, 1865)**



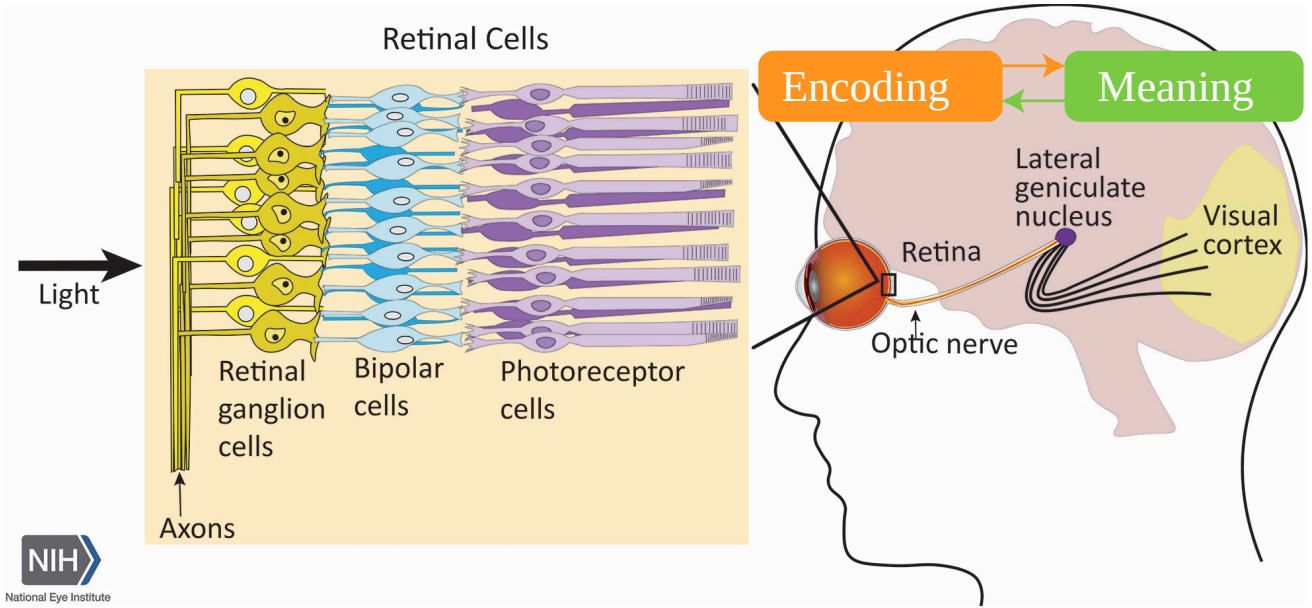

**Our eyes are not objective, they capture an edge-enhanced representation of the world.**

Shapley, R. M., & Tolhurst, D. J. (1973). Edge detectors in human vision. *Journal of Physiology*, 229, 165-183.

# Selective Vision: The Monkey Business Illusion

Simons, D. J. (2010). Monkeying around with the gorillas in our midst: Familiarity with an inattentional-blindness task does not improve the detection of unexpected events. *i-Perception*. https://doi.org/10.1068/i0386

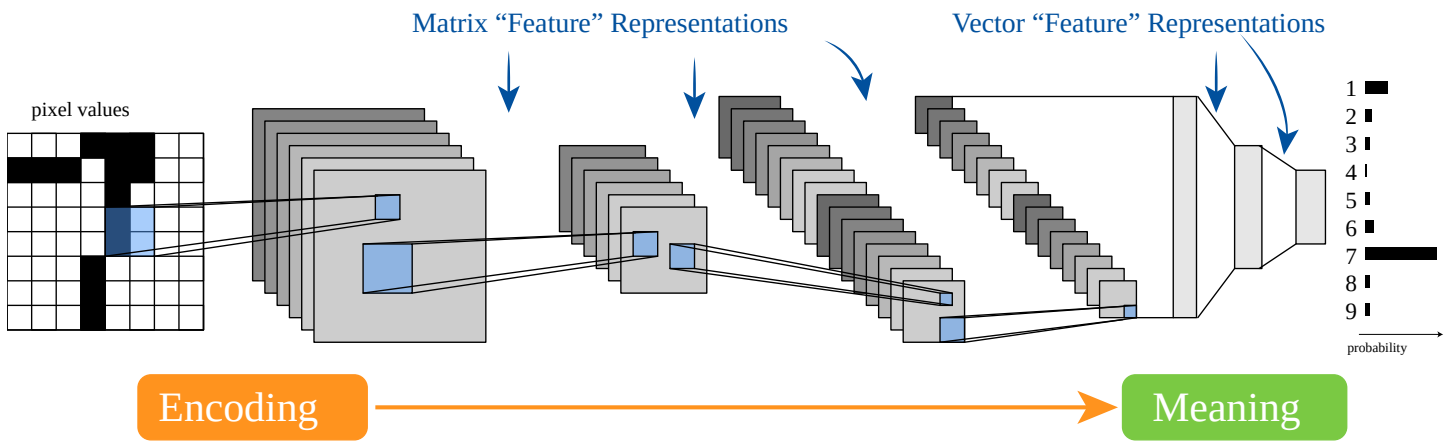
# Our mind selects what we perceive




Yang, J., et al. (2025). Differential impact of retinal lesions on visual responses of LGN X and Y cells. *The Journal of Neuroscience*. DOI: 10.1523/JNEUROSCI.0436-25.2025.

# From Encoding to Meaning

## Feature Learning in Neural Networks



# Excurse: Dimensionality Reduction

We often need to visualize high-dimensional data on a lower-dimensional space, as here with 3D data on a 2D screen.

**Principal Component Analysis (PCA)** linearly rotates high-dimensional data points so that a 2D projection best captures the variance (i.e., spread) of the data

Rotate to PCA projection | Reset | ☑ Show projections

**t-distributed Stochastic Neighborhood Embedding (t-SNE)** moves data points onto a 2D plane and tries to preserve the local neighborhood relation between points.

Run t-SNE projection | Reset | ☑ Show trajectories

# Feature Representations in LeNet (2000)

# Supervised Learning: From Labels to Loss

## Learning from labeled images

- **Input:** a 28×28 image of a handwritten digit
- **Known truth:** the human-provided label `7`
- **Prediction:** probabilities over the ten digit classes
- **Learning:** iteratively adjust the model to minimize the prediction error
- **Goal:** make the probability for the true label `7` as high as possible

`image pixels → CNN representation → class probabilities → label`

> **How do we measure whether the prediction is correct or incorrect?**

## Classification example: Cross-Entropy Loss

$$\mathcal{L}_{CE}(y, \hat{y}) = -\sum_{c=1}^{C} y_c \log(\hat{y}_c)$$

For a one-hot target label:

$$\mathcal{L}_{CE} = -\log(\hat{y}_{\text{correct}})$$

Correct class: `7`

- If `p(7) = 0.90` → low loss
- If `p(7) = 0.10` → high loss

> **Cross entropy rewards high probability on the correct class and penalizes confidence in wrong classes.**

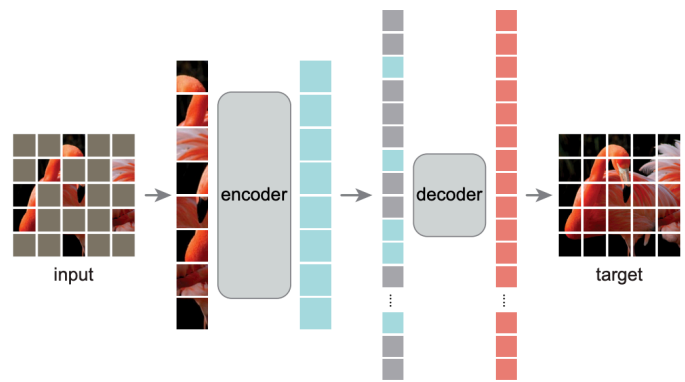
# Self-supervised Learning

## Contrastive Learning

Identify the corresponding object among two different views of the same object.

## Masked Autoencoding

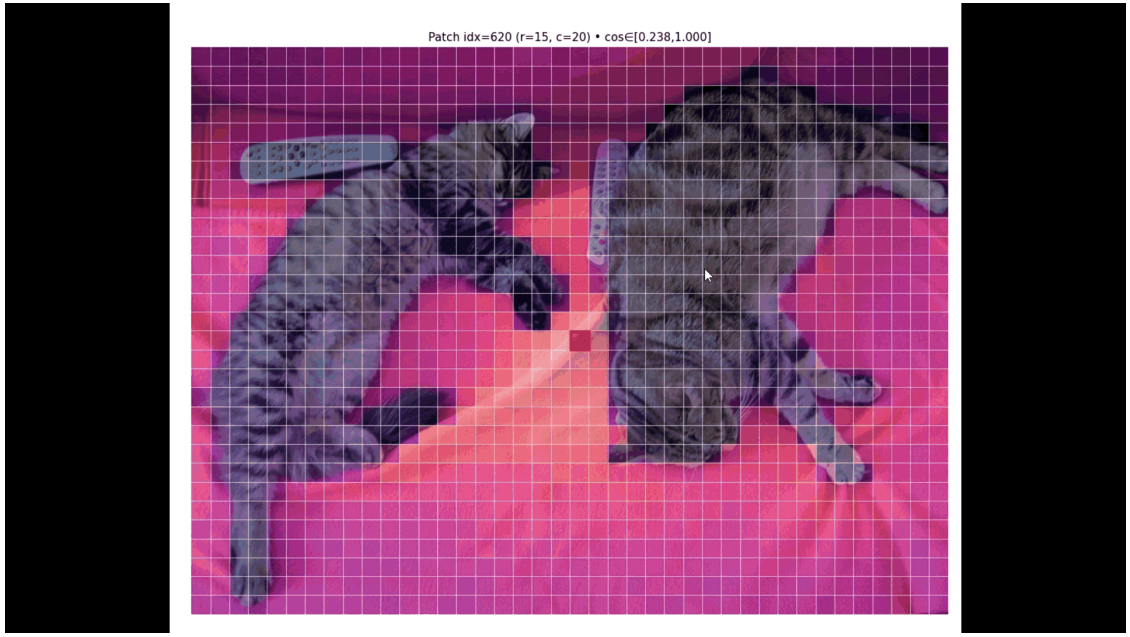
Train the model to correctly inpaint a masked portion of an image.




Caron, M., Misra, I., Mairal, J., Goyal, P., Bojanowski, P., & Joulin, A. (2020). **Unsupervised Learning of Visual Features by Contrasting Cluster Assignments** . NeurIPS. Code: **facebookresearch/swav**.

He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2022). **Masked autoencoders are scalable vision learners** . CVPR, 16000-16009.

# High-level Visual Features (Dinov3, 2025)



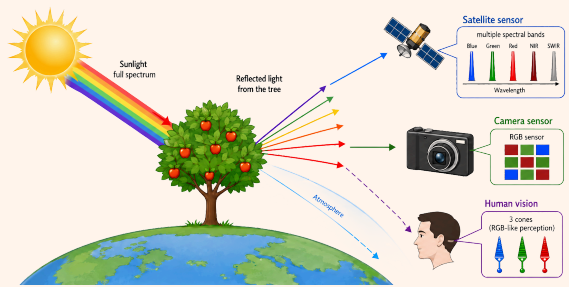

Demo: **devMuniz02/DINOv3-Interactive-Patch-Cosine-Similarity**
DINOv3 model: **Siméoni, O., Vo, H. V., Seitzer, M., Baldassarre, F., Oquab, M., Jose, C., Khalidov, V., Szafraniec, M., Yi, S., Ramamonjisoa, M., et al. (2025). DINOv3. *arXiv:2508.10104*.**

# High-level Visual Features (SAM 2, 2024)

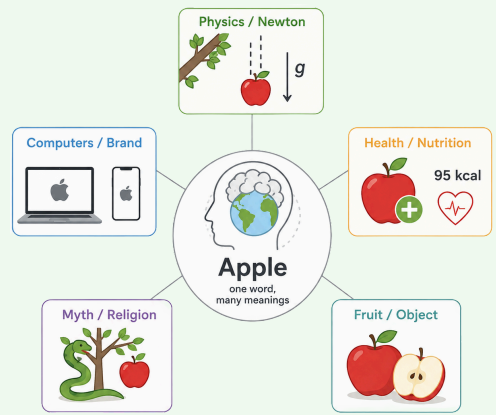
Try the Segment Anything Model (SAM) 2 demo

# From Physical to Societal Representations

## Observing the Physical World with Vision




## Understanding Societies by Language

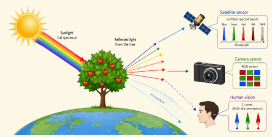


# Lecture Outline

## Representations

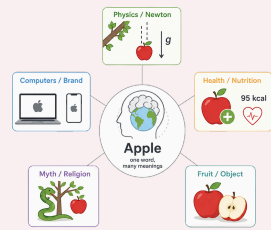
What are representations?

## Vision



How vision encodes the world

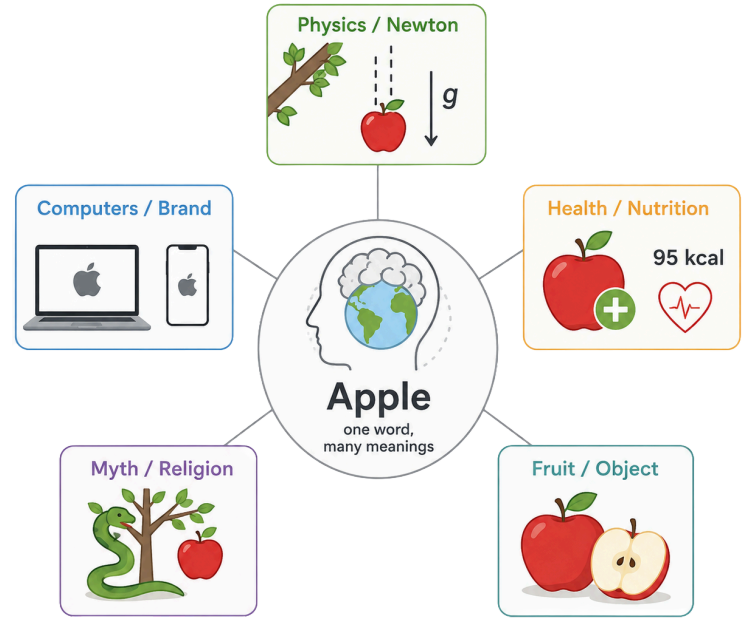
## Language



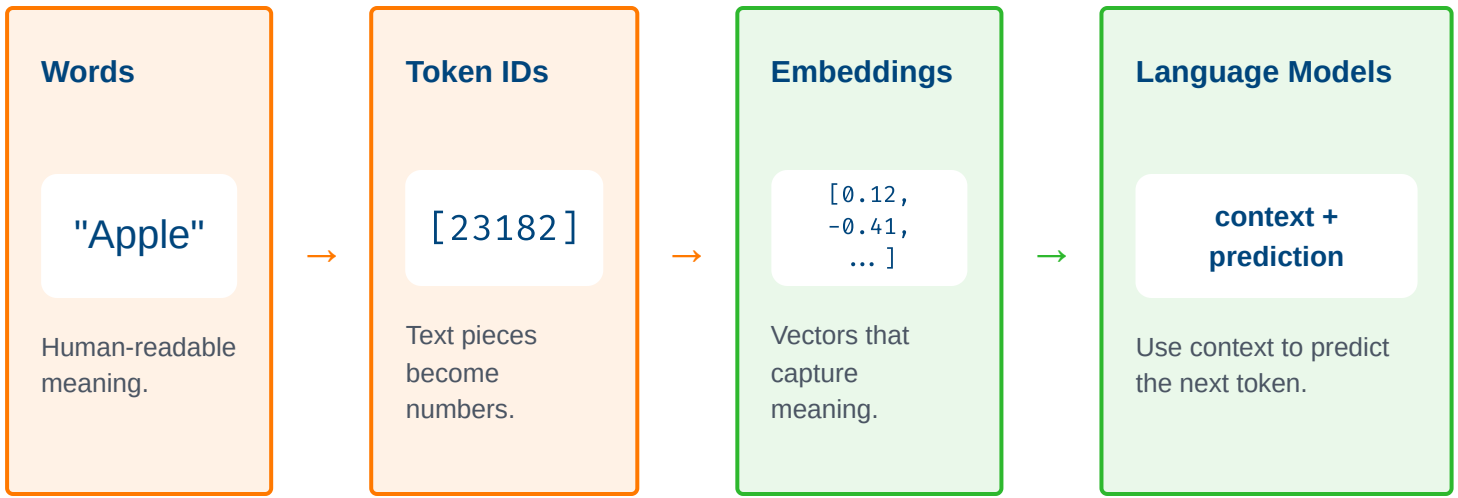

How text becomes meaning

Die Stadt in meinem Kopf. (2016). *Süddeutsche Zeitung*.

# Language Representations



# From Text Encoding to Language Models



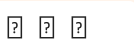

**From symbols to meaning in context.**

# Character Encodings

## Human writing systems

Symbols carry meaning.

**Cuneiform**
c. 3200 BCE



**Hieroglyphs**
c. 3100 BCE

**Chinese characters**
c. 1200 BCE - today

**Alphabetic writing**
c. 700 BCE - today

A B C ...

**Digital-era symbols**
20th-21st century

@ # :) 😂 🌍

**Learned meaning**
different fonts, same symbol

A A A A A A A A

## Machine character encodings

Computers store characters as numbers.

**ASCII**
early character numbers

```
A → 65
B → 66
a → 97
```

**Unicode code points**
global character IDs

```
A → U+0041
ä → U+00E4
🌍 →
U+1F30D
```

**UTF-8 bytes**
stored byte sequences

```
A → 41
ä → C3 A4
🌍 → F0 9F
8C 8D
```

**What does "A" + "B" give in C++? Try cpp.sh**

```
std::cout << ('A' + 'B') << std::endl;
```

# Word Tokenizer

## Language Models tokenize text

**Question: How many "r"s are in Strawberry?**

| | |
|---|---|
| **Our view** | `S t r a w b e r r y` |
| **Model view** | `[Str] [aw] [berry]` |
| **Token IDs** | `[3504, 1134, 19772]` |

Vocabulary: ~100k–200k tokens.

Try it yourself: platform.openai.com/tokenizer

Try: Strawberry · Schifffahrt · Poppelsdorfer Allee

## We also read in chunks

Hmuans do not raed ervey wrod letetr by letetr. As lnog as the frist and lsat lettres are rghit, the txet is sitll esay to raed.

We read learned patterns, not single letters.

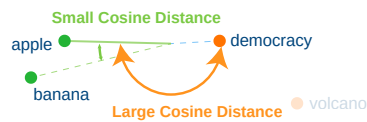
# Word Embeddings

## From token IDs to vectors

| token | ID | vector |
|---|---|---|
| strawberry | 8123 | [0.21, -0.11, ...] |
| apple | 15421 | [-0.35, 0.42, ...] |
| banana | 9821 | [0.19, 0.05, ...] |

IDs are arbitrary. Vectors are learned and capture meaning.

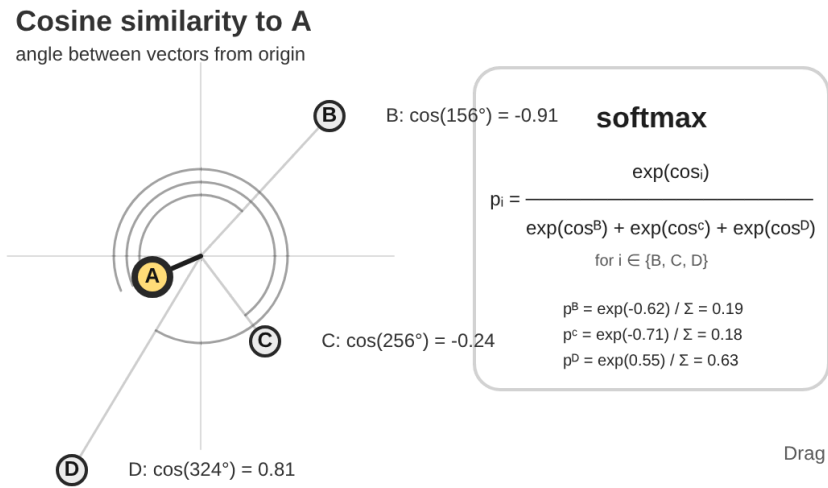
## Words are points in a semantic space

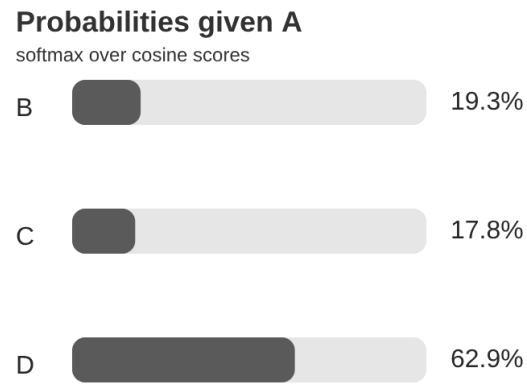



Each word is a vector.

Similarity = cosine of the angle.

# Softmax: Cosine Similarity to Probabilities




Drag A, B, C, or D. Probabilities are computed relative to A.

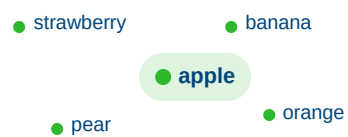
# Context matters

**Same word, different neighbours.**

## Fruit context

I ate an **apple**.

2D illustration of high-dimensional embeddings

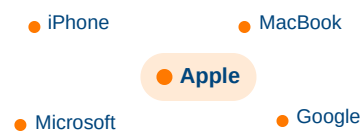



nearby words: fruit, sweet, edible

## Technology context

I bought the new **Apple**.

2D illustration of high-dimensional embeddings




nearby words: device, software, company

# Language Models predict the next token.

**Static embedding**

`apple → one vector`

**Contextual embedding**

`apple → vector depends on context`

**Inputs**

1. masked word: `<CLS>`
2. context

→

**Language model**

`p(<CLS> | context)`

→

**Output probabilities**

```
fruit: 0.42
company: 0.31
device: 0.18
other: 0.09
```

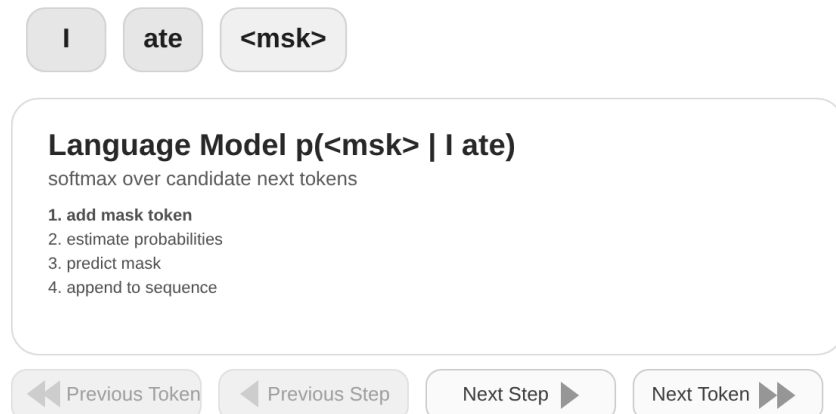
**Try it: huggingface.co/spaces/ysdede/fill-mask-demo**

# Autoregressive next token prediction

### One token at a time

Predict the next token, append it, repeat.

> **Autoregressive = one token after another.**




### Training signal

Trained to predict the actual next token (cross-entropy loss).

# Language Model Training

### Pre-training

Predict the next token on huge text corpora.

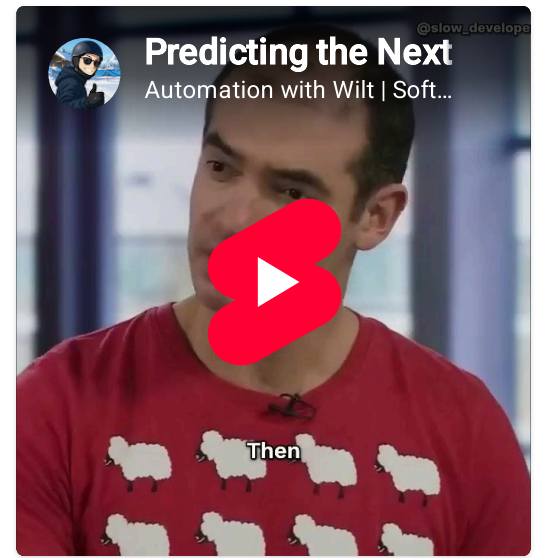
> **Predicting the next token requires world knowledge.**

### Fine-tuning

Then tuned to follow instructions.

> **→ AI assistants like ChatGPT, Claude, Mistral.**

More details: **Andrej Karpathy, “Deep Dive into LLMs like ChatGPT”**



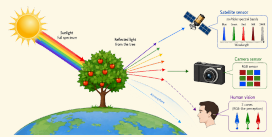

NVIDIA. (2023). *Fireside Chat With Ilya Sutskever and Jensen Huang: AI Today and Vision of the Future.* YouTube short edited by @automationwithwilt .

# Lecture Outline

## Representations

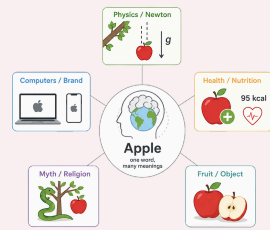
What are representations?

## Vision



How vision encodes the world

## Language



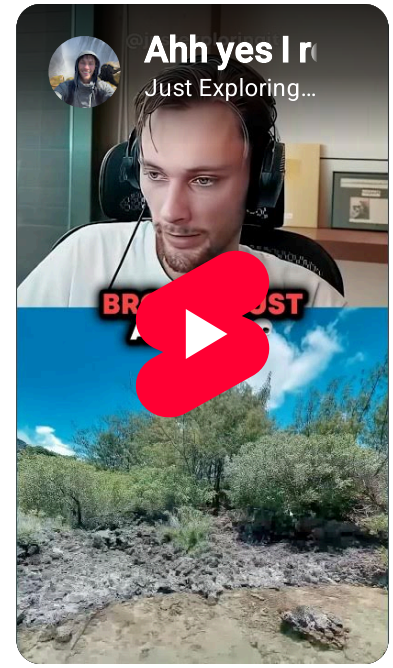

How text becomes meaning

## Spatial

How places carry spatial meaning

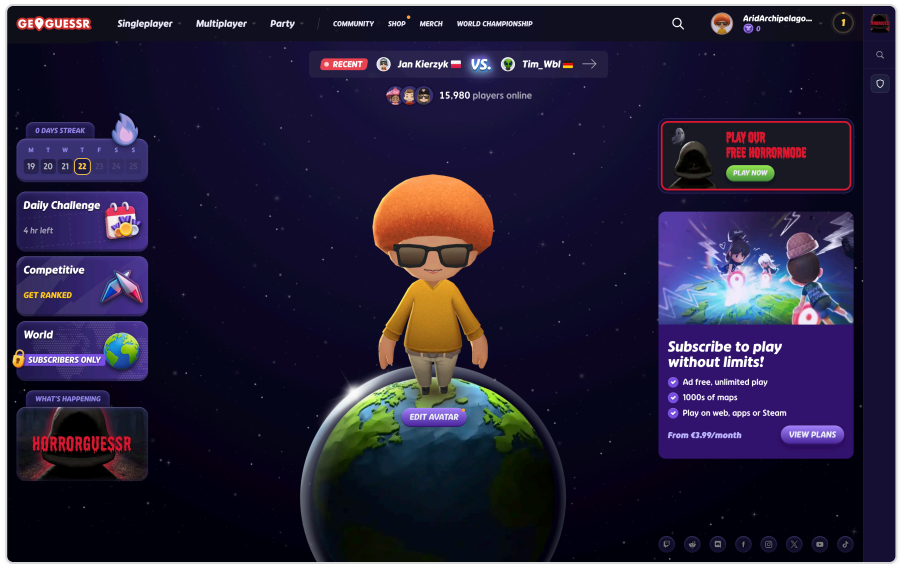
Die Stadt in meinem Kopf. (2016). *Süddeutsche Zeitung*.

# Geospatial Representations




Video: Just Exploring It, via YouTube Shorts

# GeoGuessR




📍 Play GeoGuessR

Screenshot: geoguessr.com

# Geolocalization to Learn a Mental Map

Vision

- regular city blocks,
- gabled slate roofs,
- dense housing
- irregular layout

Geo localization

- Gründerzeit, mid-19th century
- Mid-large sized city
- Remnants of a 18th
- century citadel

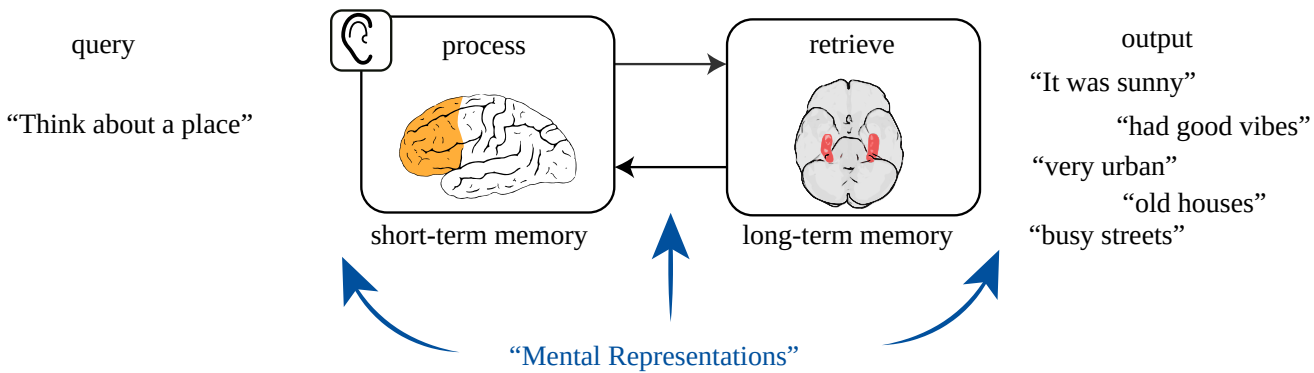
Location

(Mental) Mapping or Context

# Mental Representation of a Place

## Think of a place you visited on vacation.

Not the country or city name first — think of the actual place. What do you remember?

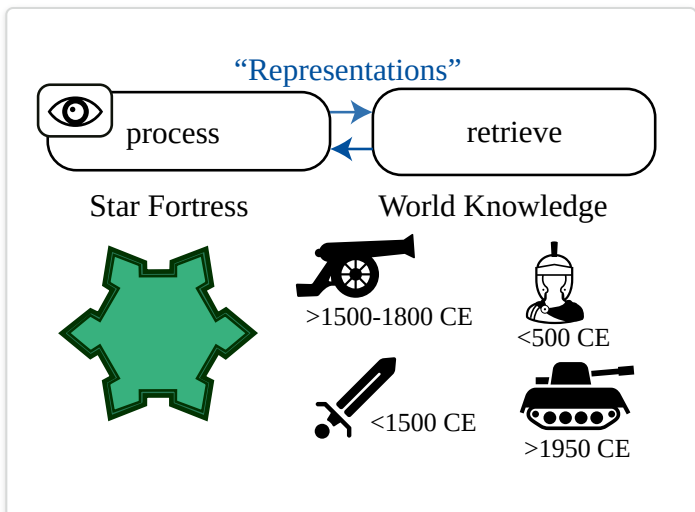



**We feel mental representations as "intuitions" and "memories" of things/places.**

Preston, A. R., & Eichenbaum, H. (2013). Interplay of hippocampus and prefrontal cortex in memory. *Current Biology*, 23(17), R764-R773.

# History: When was this city founded?

- [ ] 282 CE (Roman/Antique)
- [ ] 1153 CE (Midieval)
- [x] **1593 CE (Renaissance)**
- [ ] 1975 CE (Modern)



# Biodiversity: Which Agricultural Landscape supports more Diverse Bird Populations?

Engist, D., Finger, R., Knaus, P., Guélat, J., & Wuepper, D. (2023). Agricultural systems and biodiversity: evidence from European borders and bird populations. *Ecological Economics*, 209, 107854.

# Learning a Map Representation

**Image Encoder**

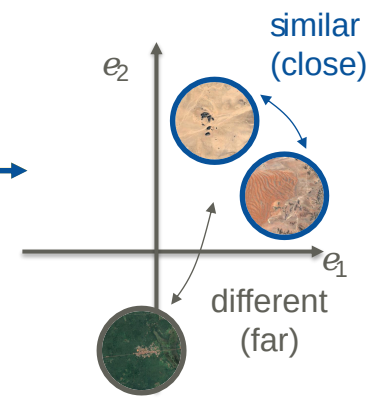
**Geospatial Earth Embeddings**

$e_1$ $e_2$ $e_3$ $e_4$ $e_5$

**Location Encoder**

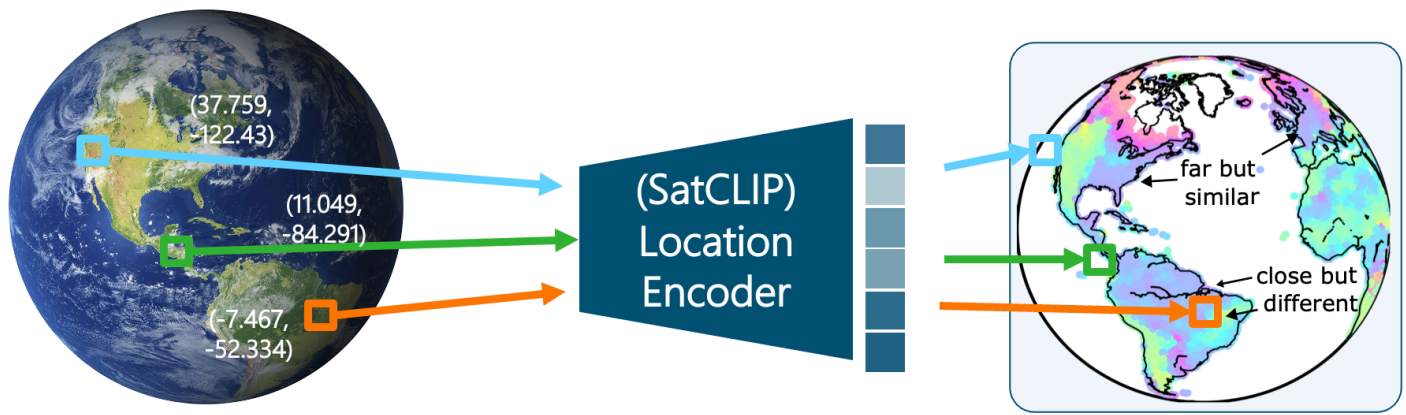
$e_1$ $e_2$ $e_3$ $e_4$ $e_5$

**Location**
(longitude, latitude, time)



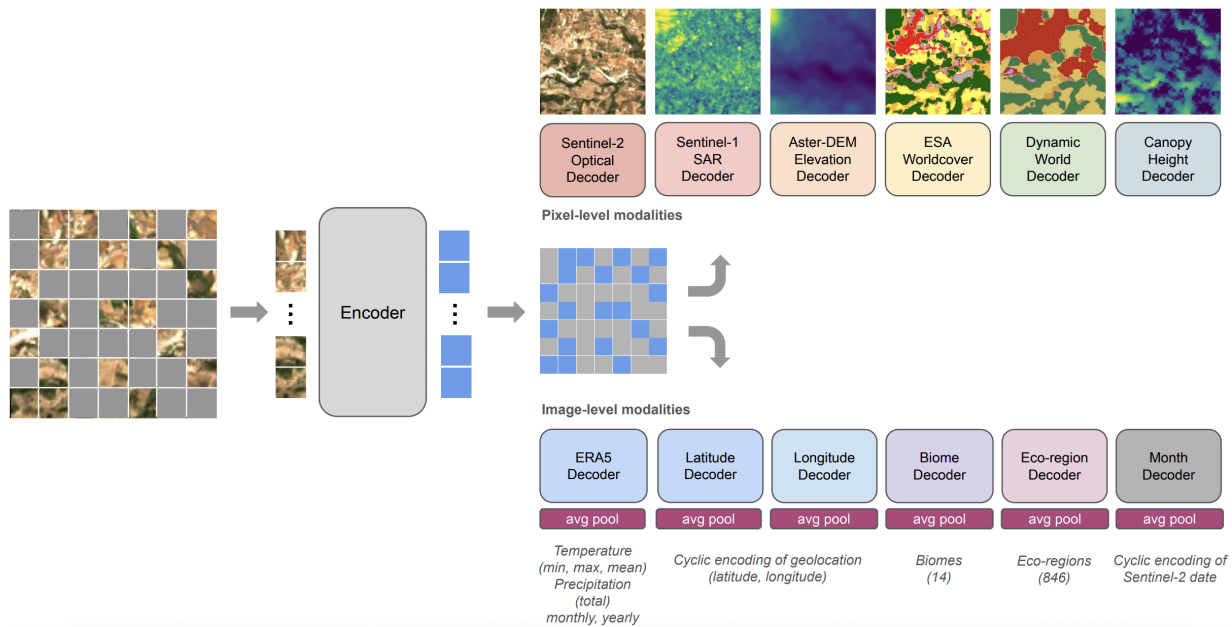
# Satellite Contrastive Location-Image Pre-training (SatCLIP)

# SatCLIP Encoder for Implicit Embeddings

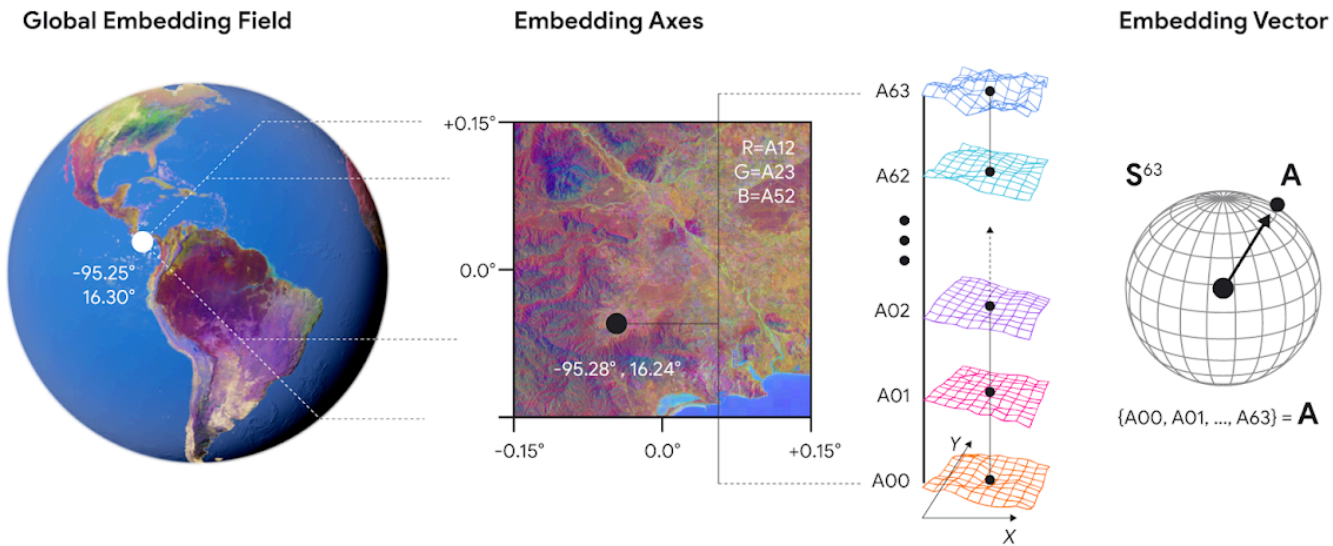


# Geospatial Foundation Models

## Encode a place through Deep Models

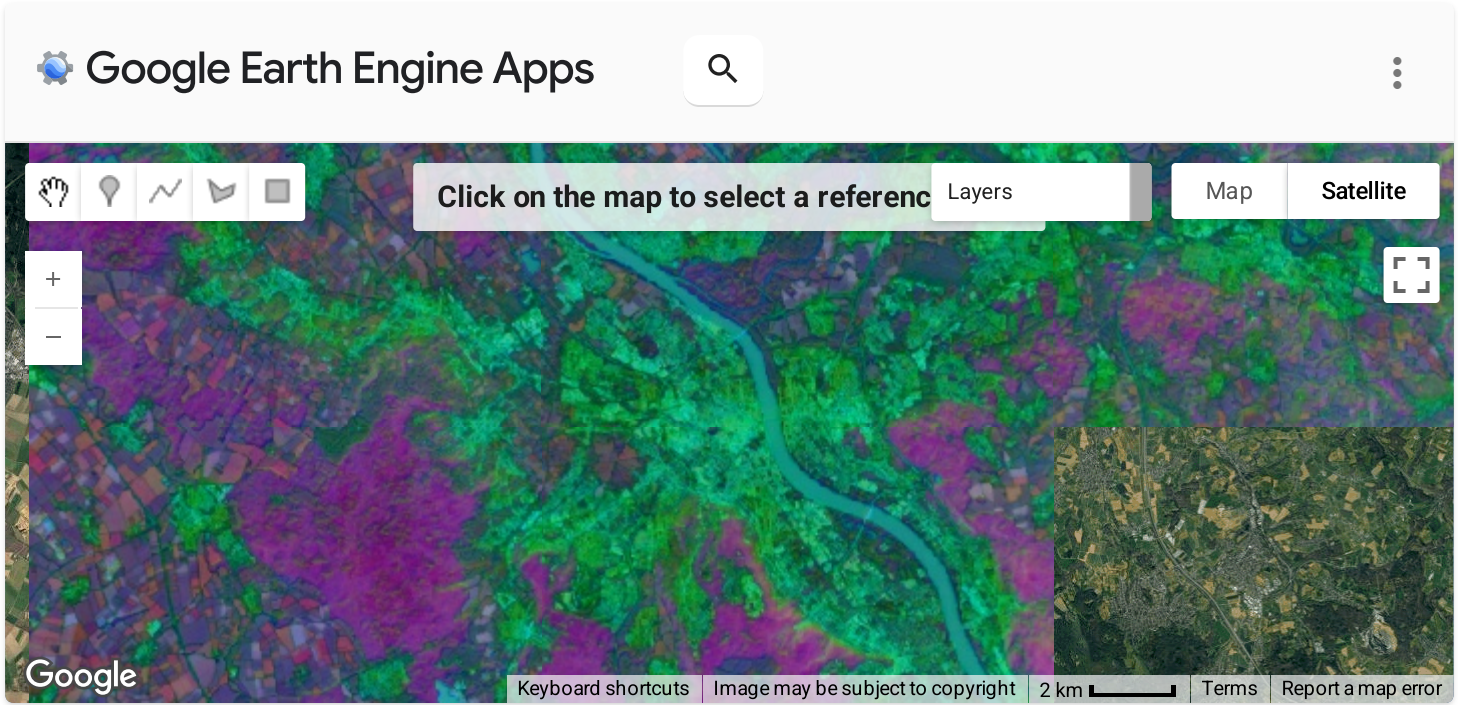


# Alpha Earth Embeddings

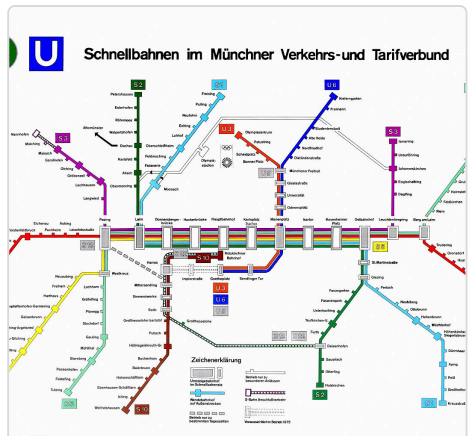



Brown, C. F., Kazmierski, M. R., Pasquarella, V. J., Rucklidge, W. J., Samsikova, M., Zhang, C., ... & Kohli, P. (2025). Alphaearth foundations: An embedding field model for accurate and efficient global mapping from sparse label data. *arXiv preprint arXiv:2507.22291*.

# Alpha Earth Embedding Explorer

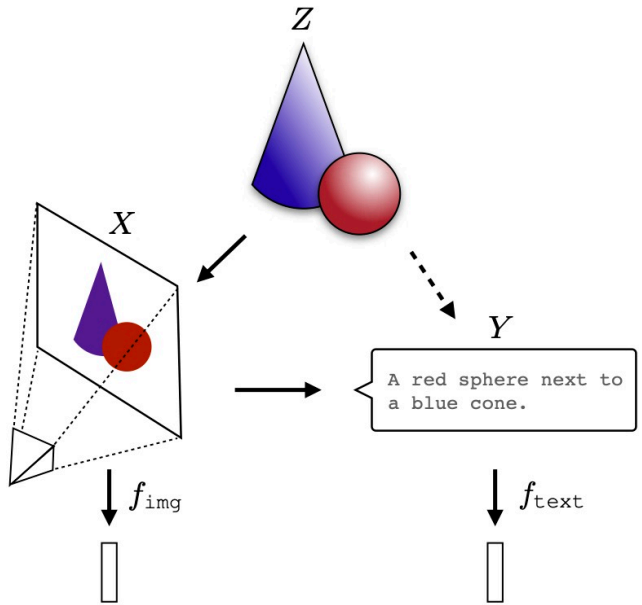


# Takeaways - Same city, different maps



We need to select or learn a suitable representation for the problem-at-hand.

# The Platonic Representation Hypothesis




**Neural networks, trained with different objectives on different data and modalities, are converging to a shared statistical model of reality in their representation spaces.**

Huh, M., Cheung, B., Wang, T., & Isola, P. (2024). *The Platonic Representation Hypothesis*. Position paper, ICML 2024.

# Revisit the learning outcomes

Light gray fill: your group's before responses today · Click to respond again.

**Discuss** representations of the same object and **choose** a suitable one for a problem.

Not comfortable — Very comfortable

Click to respond

**Discuss** the similarities and differences in visual, word, or geospatial representations.

Not comfortable — Very comfortable

Click to respond

**Understand** how cosine similarity measures how close two embeddings are.

Not comfortable — Very comfortable

Click to respond

**Understand** how PCA and t-SNE visualize high-dimensional vector representations.

Not comfortable — Very comfortable

Click to respond

**Describe** how tokenization, embeddings, and next-token prediction represent language.

Not comfortable — Very comfortable

Click to respond

**Understand** what geospatial information is encoded in geospatial embeddings.

Not comfortable — Very comfortable

Click to respond

# See you at 13:00 for Lab 1!

## Visualizing high-dimensional representation spaces

We will investigate how **PCA** and **t-SNE** reveal the structure of embeddings.

**13:00–15:00 · Seminarraum 2.003**
Nußallee 1